

PBMC 3K / 6K / 10K Performance Profiling

Scope and method

Single-threaded Scanpy 1.10.2 workflow (PBMC Matrix Market input); one measured run per dataset. The reader cache is disabled so input timing is comparable. Each run uses `/usr/bin/time -v` for process-level resources and `time.perf_counter` around every workflow section. `rank_gene_groups` is executed exactly once, under cProfile.

Key findings

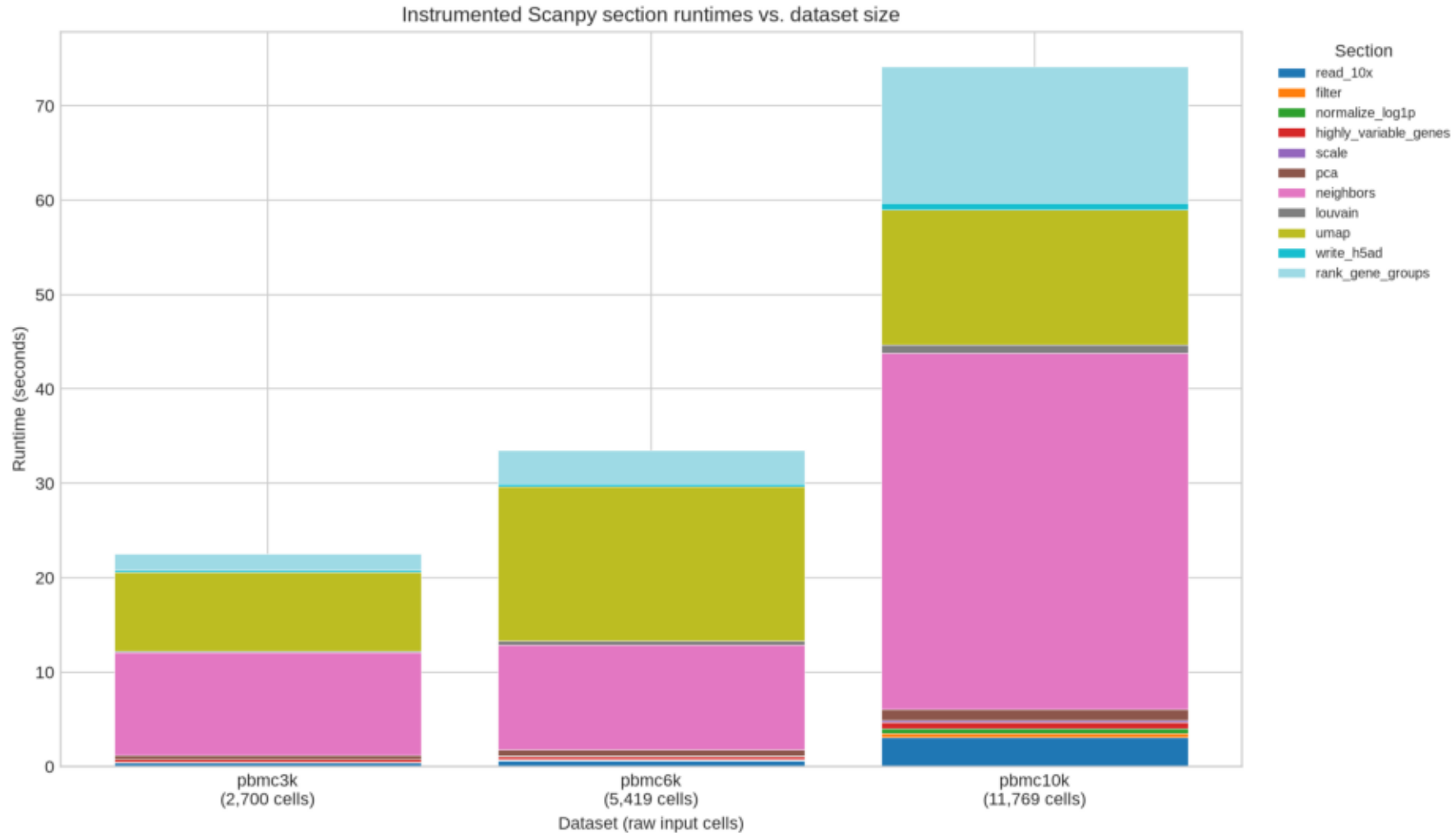
- Total instrumented runtime grows from 22.5s (3K) to 74.1s (10K); external wall time is 25.26s to 77.23s.
- At 10K, `neighbors` is the largest coarse section (37.7s), followed by `rank_gene_groups` (14.4s) and `UMAP` (14.3s).
- Within `rank_gene_groups`, cProfile identifies `pandas rank / Scanpy _ranks` as dominant; sparse column indexing and basic-statistics work become substantial at 10K.
- Peak RSS increases from 0.82 GiB (3K) to 1.66 GiB (10K), consistent with dense scaling and larger graph/embedding data.

20K estimate

The projection is an ordinary least-squares linear fit of the three observed points against raw cell count. It is a planning estimate, not a guarantee: neighbor graph construction and UMAP can scale nonlinearly, and cluster composition changes rank-gene work.

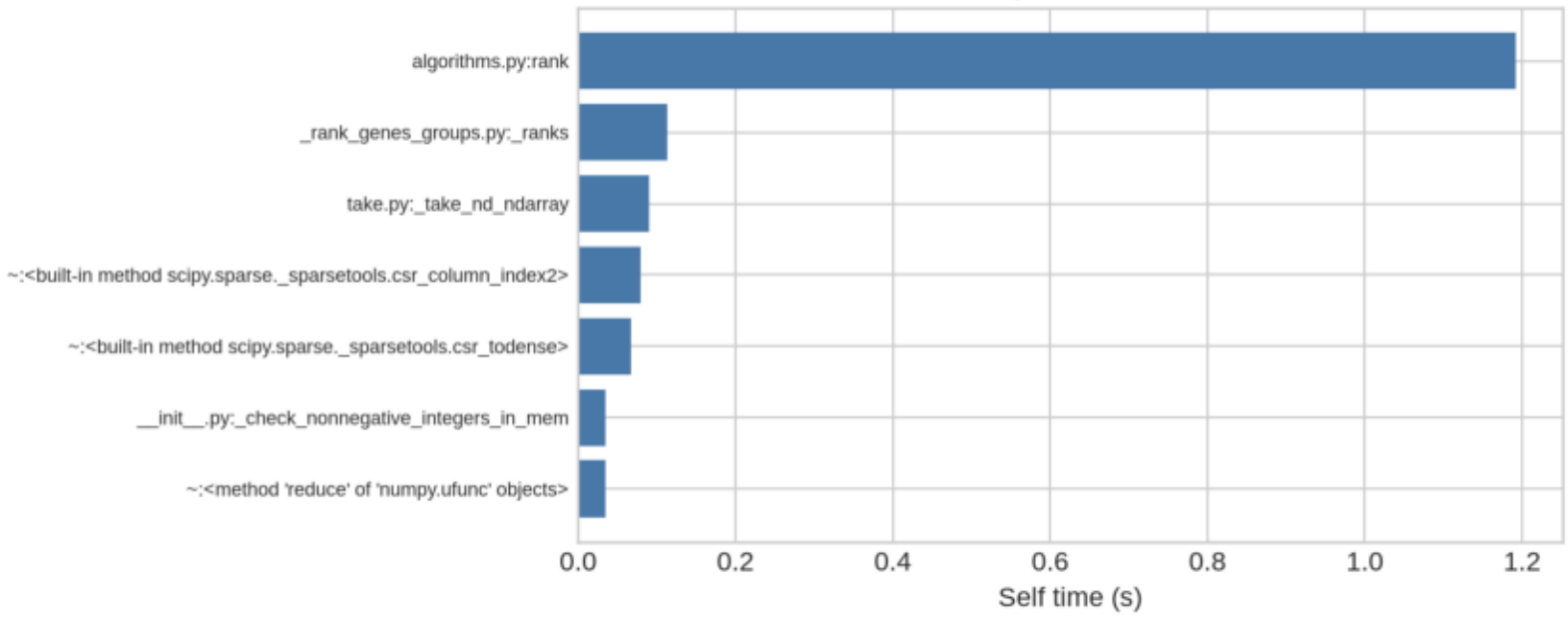
Coarse-grain wall time, CPU utilization, and memory

Dataset	Input cells	Genes	Wall (s)	CPU (%)	Peak RSS (GiB)
pbmc3k	2700	32738	25.26	97	0.82
pbmc6k	5419	32738	36.49	98	0.87
pbmc10k	11769	33538	77.23	100	1.66

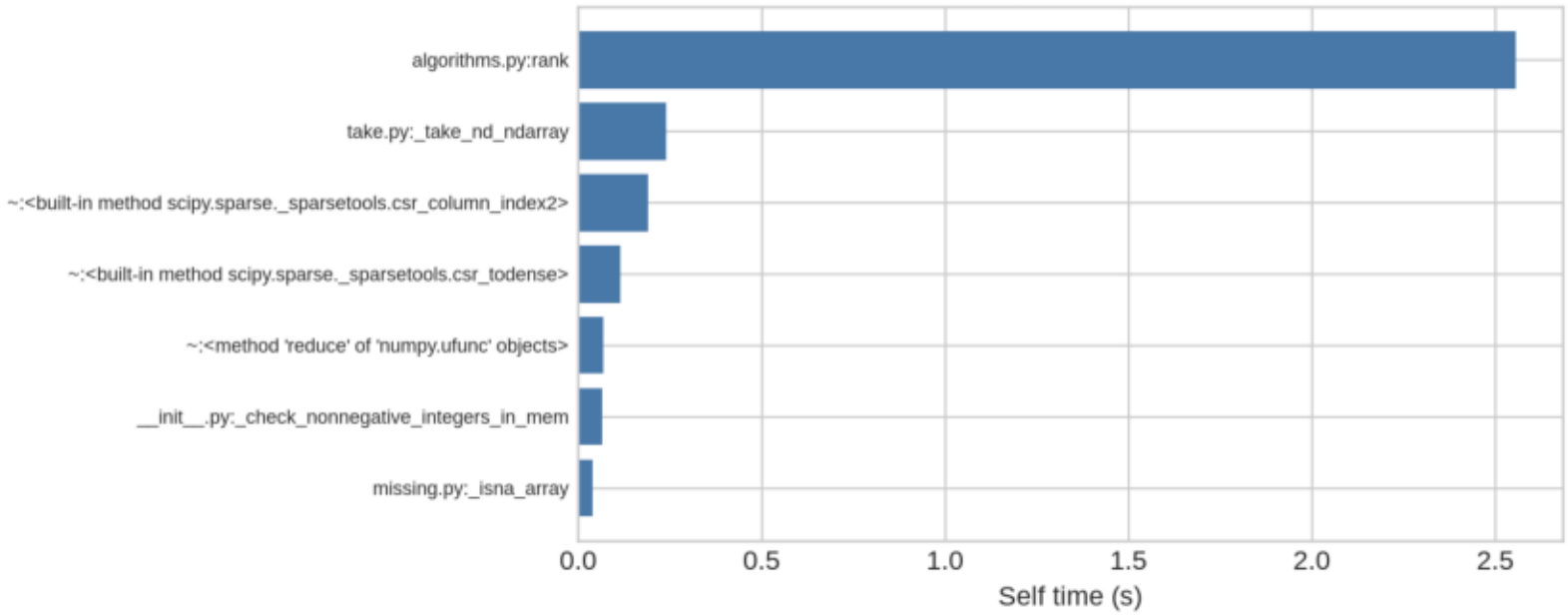


cProfile: rank_gene_groups top functions by self time

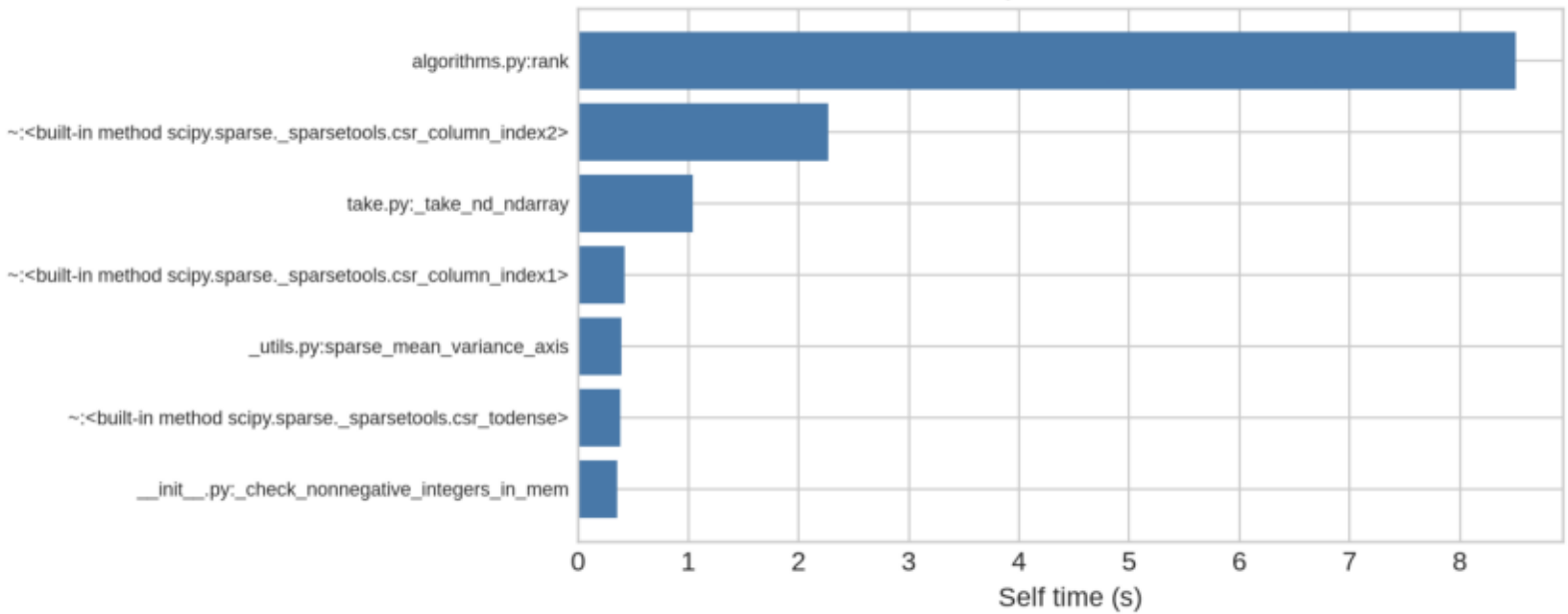
pbmc3k



pbmc6k



pbmc10k



20K extrapolation and bottleneck assessment

20K projected runtimes (seconds)

	20K projected seconds
read_10x	5.6
filter	0.8
normalize_log1p	0.8
highly_variable_genes	1.1
scale	0.3
pca	2.0
neighbors	62.4
louvain	1.5
umap	19.4
write_h5ad	1.3
rank_gene_groups	26.0
TOTAL (instrumented sections)	121.2

Interpretation: neighbors is the main projected cost (~62.4s). Rank-gene work (~26.0s) is the r hotspot. The fitted instrumented total is ~121.2s; allow additional startup/import overhead and variability.

Slurm batch scripts and reproduction

Submit commands

```
sbatch slurm/pbmc3k.sbatch  
sbatch slurm/pbmc6k.sbatch  
sbatch slurm/pbmc10k.sbatch
```

Batch configuration

```
pbmc3k: 1 CPU, 16G, 01:00:00  
pbmc6k: 1 CPU, 24G, 02:00:00  
pbmc10k: 1 CPU, 32G, 04:00:00
```

Each script changes to `SLURM_SUBMIT_DIR` and invokes `scripts/run_pbmc_experiment.sh`. The runner `bmi500` environment, constrains BLAS/OMP/Numba to the allocated CPU count, writes `logs/<dataset>` results/`<dataset>.scanpy.h5ad`, and `profiles/rank_genes_<dataset>_profile.prof`.

Reproducibility note

The figure and PDF are regenerated with: `python generate_report.py`

Appendix: Slurm batch-script listings

```
--- slurm/pbmc3k.sbatch ---
#!/usr/bin/env bash
#SBATCH --job-name=pbmc3k-profile
#SBATCH --output=slurm-pbmc3k-%j.out
#SBATCH --error=slurm-pbmc3k-%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=16G
#SBATCH --time=01:00:00

set -euo pipefail
cd "${SLURM_SUBMIT_DIR}"
bash scripts/run_pbmc_experiment.sh pbmc3k "${SLURM_CPUS_PER_TASK}"

--- slurm/pbmc6k.sbatch ---
#!/usr/bin/env bash
#SBATCH --job-name=pbmc6k-profile
#SBATCH --output=slurm-pbmc6k-%j.out
#SBATCH --error=slurm-pbmc6k-%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=24G
#SBATCH --time=02:00:00

set -euo pipefail
cd "${SLURM_SUBMIT_DIR}"
bash scripts/run_pbmc_experiment.sh pbmc6k "${SLURM_CPUS_PER_TASK}"

--- slurm/pbmc10k.sbatch ---
#!/usr/bin/env bash
#SBATCH --job-name=pbmc10k-profile
#SBATCH --output=slurm-pbmc10k-%j.out
#SBATCH --error=slurm-pbmc10k-%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=32G
#SBATCH --time=04:00:00

set -euo pipefail
cd "${SLURM_SUBMIT_DIR}"
bash scripts/run_pbmc_experiment.sh pbmc10k "${SLURM_CPUS_PER_TASK}"
```

Appendix: pbmc3k console output (1–57 of 104)

```
===== 2026-09-14T15:25:50-04:00 pbmc3k =====
host=bignode2.priv.bmi.emory.edu threads=1
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/_utils/__init__.py:2
7: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_csv from `anndata` is deprecated. Import anndata.io.read_csv instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_excel from `anndata` is deprecated. Import anndata.io.read_excel instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_hdf from `anndata` is deprecated. Import anndata.io.read_hdf instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_loom from `anndata` is deprecated. Import anndata.io.read_loom instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_mtx from `anndata` is deprecated. Import anndata.io.read_mtx instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_text from `anndata` is deprecated. Import anndata.io.read_text instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/__init__.py:70:
FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import anndata.io.read_umi_tools instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/louvain/__init__.py:54:
UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html.
The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin
to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
RUN_METADATA dataset=pbmc3k threads=1
SECTION_TIMING dataset=pbmc3k section=read_10x seconds=0.3717
DATASET_SHAPE dataset=pbmc3k cells=2700 genes=32738
filtered out 19024 genes that are detected in less than 3 cells
SECTION_TIMING dataset=pbmc3k section=filter seconds=0.0486
normalizing counts per cell
  finished (0:00:00)
SECTION_TIMING dataset=pbmc3k section=normalize_log1p seconds=0.0488
extracting highly variable genes
  finished (0:00:00)
SECTION_TIMING dataset=pbmc3k section=highly_variable_genes seconds=0.2736
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale
.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
SECTION_TIMING dataset=pbmc3k section=scale seconds=0.0431
computing PCA
  with n_comps=30
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.p
y:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad._version__) < Version("0.9")
  finished (0:00:00)
SECTION_TIMING dataset=pbmc3k section=pca seconds=0.3965
computing neighbors
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/__init__.p
y:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of
```

Appendix: pbmc3k console output (58–104 of 104)

```
obsrn_keys. (e.g. `k` in `adata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" in adata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished (0:00:10)
SECTION_TIMING dataset=pbmc3k section=neighbors seconds=10.9199
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished (0:00:00)
SECTION_TIMING dataset=pbmc3k section=louvain seconds=0.1349
computing UMAP
  finished (0:00:08)
SECTION_TIMING dataset=pbmc3k section=umap seconds=8.3631
SECTION_TIMING dataset=pbmc3k section=write_h5ad seconds=0.1747
ranking genes
  finished (0:00:01)
SECTION_TIMING dataset=pbmc3k section=rank_gene_groups seconds=1.7605
TIMINGS_JSON {"dataset": "pbmc3k", "sections": {"filter": 0.04861556878313422, "highly_variable_genes":
0.2736478568986058, "louvain": 0.1349168261513114, "neighbors": 10.919944623019546, "normalize_log1p":
0.04876408586278558, "pca": 0.3965183920226991, "rank_gene_groups": 1.7605236792005599, "read_10x":
0.37174492981284857, "scale": 0.04309308901429176, "umap": 8.363101454917341, "write_h5ad": 0.1747362487949431}}
PROFILE_OUTPUT /home/swen30/Desktop/bmi500-profiling/profiles/rank_genes_pbmc3k_profile.prof
  Command being timed: "python /home/swen30/Desktop/bmi500-profiling/scanpy_pbmc.py --data-dir
/home/swen30/Desktop/bmi500-profiling/data --data-set pbmc3k --out-dir
/home/swen30/Desktop/bmi500-profiling/results --profile-dir /home/swen30/Desktop/bmi500-profiling/profiles
--num-threads 1"
  User time (seconds): 23.83
  System time (seconds): 0.91
  Percent of CPU this job got: 97%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:25.26
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 856996
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 0
  Minor (reclaiming a frame) page faults: 169505
  Voluntary context switches: 5828
  Involuntary context switches: 141
  Swaps: 0
  File system inputs: 0
  File system outputs: 86640
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

Appendix: pbmc6k console output (1–57 of 105)

```
==== 2026-09-14T15:26:19-04:00 pbmc6k ====
host=bignode2.priv.bmi.emory.edu threads=1
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/_utils/_init__.py:2
7: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_csv from `anndata` is deprecated. Import anndata.io.read_csv instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_excel from `anndata` is deprecated. Import anndata.io.read_excel instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_hdf from `anndata` is deprecated. Import anndata.io.read_hdf instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_loom from `anndata` is deprecated. Import anndata.io.read_loom instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_mtx from `anndata` is deprecated. Import anndata.io.read_mtx instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_text from `anndata` is deprecated. Import anndata.io.read_text instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import anndata.io.read_umi_tools instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/louvain/_init__.py:54:
UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html.
The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin
to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
RUN_METADATA dataset=pbmc6k threads=1
SECTION_TIMING dataset=pbmc6k section=read_10x seconds=0.5847
DATASET_SHAPE dataset=pbmc6k cells=5419 genes=32738
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
SECTION_TIMING dataset=pbmc6k section=filter seconds=0.0693
normalizing counts per cell
  finished (0:00:00)
SECTION_TIMING dataset=pbmc6k section=normalize_log1p seconds=0.0768
extracting highly variable genes
  finished (0:00:00)
SECTION_TIMING dataset=pbmc6k section=highly_variable_genes seconds=0.3014
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale
.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
SECTION_TIMING dataset=pbmc6k section=scale seconds=0.0749
computing PCA
  with n_comps=30
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.p
y:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:00)
SECTION_TIMING dataset=pbmc6k section=pca seconds=0.6829
computing neighbors
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/_init__.p
```

Appendix: pbmc6k console output (58–105 of 105)

```
y:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of
obsm_keys. (e.g. `k inadata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" inadata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished (0:00:11)
SECTION_TIMING dataset=pbmc6k section=neighbors seconds=11.0895
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished (0:00:00)
SECTION_TIMING dataset=pbmc6k section=louvain seconds=0.3982
computing UMAP
  finished (0:00:16)
SECTION_TIMING dataset=pbmc6k section=umap seconds=16.3776
SECTION_TIMING dataset=pbmc6k section=write_h5ad seconds=0.2813
ranking genes
  finished (0:00:03)
SECTION_TIMING dataset=pbmc6k section=rank_gene_groups seconds=3.5404
TIMINGS_JSON {"dataset": "pbmc6k", "sections": {"filter": 0.06929873721674085, "highly_variable_genes":
0.30136993015185, "louvain": 0.39824601309373975, "neighbors": 11.089493577834219, "normalize_log1p":
0.0768445716239512, "pca": 0.682878389954567, "rank_gene_groups": 3.540375758893788, "read_10x":
0.58472122810781, "scale": 0.07489694515243173, "umap": 16.377572338096797, "write_h5ad": 0.28132627671584487}}
PROFILE_OUTPUT /home/swen30/Desktop/bmi500-profiling/profiles/rank_genes_pbmc6k_profile.prof
  Command being timed: "python /home/swen30/Desktop/bmi500-profiling/scanpy_pbmc.py --data-dir
/home/swen30/Desktop/bmi500-profiling/data --data-set pbmc6k --out-dir
/home/swen30/Desktop/bmi500-profiling/results --profile-dir /home/swen30/Desktop/bmi500-profiling/profiles
--num-threads 1"
    User time (seconds): 34.55
    System time (seconds): 1.23
    Percent of CPU this job got: 98%
    Elapsed (wall clock) time (h:mm:ss or m:ss): 0:36.49
    Average shared text size (kbytes): 0
    Average unshared data size (kbytes): 0
    Average stack size (kbytes): 0
    Average total size (kbytes): 0
    Maximum resident set size (kbytes): 911328
    Average resident set size (kbytes): 0
    Major (requiring I/O) page faults: 0
    Minor (reclaiming a frame) page faults: 207704
    Voluntary context switches: 8831
    Involuntary context switches: 154
    Swaps: 0
    File system inputs: 0
    File system outputs: 160352
    Socket messages sent: 0
    Socket messages received: 0
    Signals delivered: 0
    Page size (bytes): 4096
    Exit status: 0
```

Appendix: pbmc10k console output (1–57 of 105)

```
===== 2026-09-14T15:27:12-04:00 pbmc10k =====
host=bignode2.priv.bmi.emory.edu threads=1
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/_utils/_init__.py:2
7: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_csv from `anndata` is deprecated. Import anndata.io.read_csv instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_excel from `anndata` is deprecated. Import anndata.io.read_excel instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_hdf from `anndata` is deprecated. Import anndata.io.read_hdf instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_loom from `anndata` is deprecated. Import anndata.io.read_loom instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_mtx from `anndata` is deprecated. Import anndata.io.read_mtx instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_text from `anndata` is deprecated. Import anndata.io.read_text instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/anndata/_init__.py:70:
FutureWarning: Importing read_umi_tools from `anndata` is deprecated. Import anndata.io.read_umi_tools instead.
  return module.get_attr_redirect(attr_name, deprecated_mapping= DEPRECATED)
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/louvain/_init__.py:54:
UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html.
The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin
to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
RUN_METADATA dataset=pbmc10k threads=1
SECTION_TIMING dataset=pbmc10k section=read 10x seconds=3.0982
DATASET_SHAPE dataset=pbmc10k cells=11769 genes=33538
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
SECTION_TIMING dataset=pbmc10k section=filter seconds=0.4297
normalizing counts per cell
  finished (0:00:00)
SECTION_TIMING dataset=pbmc10k section=normalize_log1p seconds=0.4630
extracting highly variable genes
  finished (0:00:00)
SECTION_TIMING dataset=pbmc10k section=highly_variable_genes seconds=0.6881
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_scale
.py:316: UserWarning: Received a view of an AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
SECTION_TIMING dataset=pbmc10k section=scale seconds=0.1780
computing PCA
  with n_comps=30
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/preprocessing/_pca.p
y:208: FutureWarning: `__version__` is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
  finished (0:00:01)
SECTION_TIMING dataset=pbmc10k section=pca seconds=1.2246
computing neighbors
/opt/scratchspace/shenghan/miniconda3_clean/envs/bmi500/lib/python3.12/site-packages/scanpy/neighbors/_init__.p
```

Appendix: pbmc10k console output (58–105 of 105)

```
y:411: FutureWarning: The method obsm_keys is deprecated and will be removed in the future. Use obsm instead of
obsm_keys. (e.g. `k inadata.obsm` or `adata.obsm.keys() | {'u'}`)
  if "X_diffmap" inadata.obsm_keys():
    using 'X_pca' with n_pcs = 30
    finished (0:00:37)
SECTION_TIMING dataset=pbmc10k section=neighbors seconds=37.7457
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished (0:00:00)
SECTION_TIMING dataset=pbmc10k section=louvain seconds=0.8591
computing UMAP
  finished (0:00:14)
SECTION_TIMING dataset=pbmc10k section=umap seconds=14.2824
SECTION_TIMING dataset=pbmc10k section=write_h5ad seconds=0.7460
ranking genes
  finished (0:00:14)
SECTION_TIMING dataset=pbmc10k section=rank_gene_groups seconds=14.4182
TIMINGS_JSON {"dataset": "pbmc10k", "sections": {"filter": 0.4296668479219079, "highly_variable_genes":
0.6881382400169969, "louvain": 0.859103660992084, "neighbors": 37.74572621891275, "normalize_log1p":
0.4629740063101053, "pca": 1.224572362843901, "rank_gene_groups": 14.418163917027414, "read_10x":
3.098241299856454, "scale": 0.17799547826871276, "umap": 14.282395214773715, "write_h5ad": 0.745970455929637}}
PROFILE_OUTPUT /home/swen30/Desktop/bmi500-profiling/profiles/rank_genes_pbmc10k_profile.prof
  Command being timed: "python /home/swen30/Desktop/bmi500-profiling/scanpy_pbmc.py --data-dir
/home/swen30/Desktop/bmi500-profiling/data --data-set pbmc10k --out-dir
/home/swen30/Desktop/bmi500-profiling/results --profile-dir /home/swen30/Desktop/bmi500-profiling/profiles
--num-threads 1"
    User time (seconds): 73.08
    System time (seconds): 4.30
    Percent of CPU this job got: 100%
    Elapsed (wall clock) time (h:mm:ss or m:ss): 1:17.23
    Average shared text size (kbytes): 0
    Average unshared data size (kbytes): 0
    Average stack size (kbytes): 0
    Average total size (kbytes): 0
    Maximum resident set size (kbytes): 1745516
    Average resident set size (kbytes): 0
    Major (requiring I/O) page faults: 0
    Minor (reclaiming a frame) page faults: 743540
    Voluntary context switches: 12314
    Involuntary context switches: 362
    Swaps: 0
    File system inputs: 280
    File system outputs: 589016
    Socket messages sent: 0
    Socket messages received: 0
    Signals delivered: 0
    Page size (bytes): 4096
    Exit status: 0
```